

MY FIRST FULL PORTFOLIO PROJECT BY- SQL FOR DATA ANALYST

All
About
Pizza!



**IN THIS PROJECT
THERE ARE MANY
PRACTICAL PROBLEM
WHICH IS SOLVED BY
SQL**

1) Retrieve the total number of orders placed.

CODE -

```
SELECT  
    COUNT(order_id) AS total_order  
FROM  
    ORDERS;
```

OUTPUT -

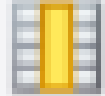
Result Grid	
	total_order
▶	21350

2) Calculate the total revenue generated from pizza sales

CODE

```
SELECT  
sum(order_details.quantity*pizzas.price) as totalsales  
FROM order_details  
JOIN pizzas  
ON pizzas.pizza_id=order_details.pizza_id;
```

OUTPUT

Result Grid			 Filter Rows:
	totalsales		
	42889.55		

3) Identify the highest-priced pizza.

CODE

```
SELECT pizza_types.name,pizzas.price  
FROM pizza_types JOIN pizzas  
ON pizza_types.pizza_type_id=pizzas.pizza_type_id  
ORDER BY pizzas.price DESC limit 1;
```

OUTPUT

Result Grid			Filter Rows:	
	name	price		
▶	The Greek Pizza	35.95		

4) Identify the most common pizza size ordered.

CODE

```
select p.size, sum(quantity) as quantity_of_most_common
from pizzas as p join order_details as od
on p.pizza_id=od.pizza_id
group by p.size order by quantity_of_most_common DESC limit 1;
```

OUTPUT

Result Grid			Filter Rows:
	size	quantity_of_most_common	
▶	L	1022	

5) List the top 5 most ordered pizza types along with their quantities.

CODE

```
SELECT pt.name,  
sum(od.quantity) as quantity  
FROM pizza_types as pt JOIN pizzas as p  
ON pt.pizza_type_id=p.pizza_type_id  
JOIN order_details as od  
ON od.pizza_id=p.pizza_id  
GROUP BY pt.name ORDER BY quantity DESC limit 5;
```

OUTPUT

Result Grid			Filter Rows:	
	name	quantity		
▶	The Pepperoni Pizza	157		
	The Barbecue Chicken Pizza	130		
	The Thai Chicken Pizza	124		
	The California Chicken Pizza	122		
	The Sicilian Pizza	117		

6) Join the necessary tables to find the total quantity of each pizza category ordered

CODE

```
SELECT pt.category,  
       sum(od.quantity) as quantity  
FROM pizza_types as pt  
JOIN pizzas as p  
ON pt.pizza_type_id=p.pizza_type_id  
JOIN order_details as od  
ON od.pizza_id=p.pizza_id  
GROUP BY pt.category ORDER BY quantity ASC;
```

OUTPUT

Result Grid





Filter Rows:

Exp

	category	quantity
▶	Chicken	573
	Veggie	613
	Supreme	637
	Classic	770

7) Determine the distribution of orders by hour of the day.

CODE

```
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

OUTPUT

Result Grid			Filter Rows:
	hour	order_count	
▶	11	1231	
	12	2520	
	13	2455	
	14	1472	
	15	1468	
	16	1920	
	17	2336	
	18	2399	
	19	2009	
	20	1642	
	21	1198	
	22	663	

8) Join relevant tables to find the category-wise distribution of pizzas.

CODE

```
SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category;
```

OUTPUT

Result Grid			Filter Rows:
	category	COUNT(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

9) Group the orders by date and calculate the average number of pizzas ordered per day.

CODE

```
SELECT
    AVG(quantity) as avg_number_pizza_per_day
FROM
    (SELECT
        o.order_date, SUM(quantity) AS quantity
    FROM
        orders AS o
    JOIN order_details AS od ON o.order_id = od.order_id
    GROUP BY o.order_date) AS orderquantity;
```

OUTPUT

Result Grid		Filter Rows:	
	avg_number_pizza_per_day		
▶	136.4737		

10) Determine the top 3 most ordered pizza types based on revenue.

CODE

```
SELECT
    pt.name, SUM(od.quantity * p.price) AS revenue
FROM
    pizza_types AS pt
    JOIN
    pizzas AS p ON pt.pizza_type_id = p.pizza_type_id
    JOIN
    order_details AS od ON od.pizza_id = p.pizza_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

OUTPUT

Result Grid			Filter Rows:	Export:
	name	revenue		
▶	The Barbecue Chicken Pizza	2317.5		
	The Thai Chicken Pizza	2293		
	The California Chicken Pizza	2099.5		

11) Calculate the percentage contribution of each pizza type to total revenue.

CODE -

```
select pt.category,round(
(sum(od.quantity*p.price) / (select
sum(od.quantity*p.price) as total_sales
from order_details as od join pizzas as p
on od.pizza_id = p.pizza_id))*100,2
)as revenve
from order_details as od join pizzas as p
on od.pizza_id=p.pizza_id
join pizza_types as pt
on pt.pizza_type_id=p.pizza_type_id
group by pt.category order by revenve DESC;
```

OUTPUT-

Result Grid			Filter Rows:
	category	revenve	
▶	Classic	26.55	
	Supreme	Supreme	
	Veggie	24.08	
	Chicken	23.82	

12) Analyze the cumulative revenue generated over time.

CODE -

```
select order_date,  
sum(revenue) over(order by order_date) as cummlative_revenue  
from  
(select o.order_date,  
sum(od.quantity*p.price) as revenue  
from order_details as od join orders as o  
on od.order_id=o.order_id  
join pizzas as p  
on p.pizza_id=od.pizza_id  
group by o.order_date) as total_sales;
```

OUTPUT-

	order_date	cummlative_revenue
►	2015-01-01	2713.8500000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.350000000002
	2015-01-11	25862.65
	2015-01-12	27721.7

13) Determine the top 3 most ordered pizza types based on revenue for each pizza category.

CODE-

```
select name, revenue, category
from
(select name ,category, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pt.category, pt.name,
sum(od.quantity*p.price) as revenue
from pizza_types as pt join pizzas as p
on pt.pizza_type_id=p.pizza_type_id
join order_details as od
on od.pizza_id=p.pizza_id
group by pt.category , pt.name ) as a) as b
where rn <=3;
```

OUTPUT -

	name	revenue	category
►	The Barbecue Chicken Pizza		Chicken
	The Thai Chicken Pizza	2293	Chicken
	The California Chicken Pizza	2099.5	Chicken
	The Pepperoni Pizza	2003.75	Classic
	The Classic Deluxe Pizza	1824.5	Classic
	The Greek Pizza	1578	Classic
	The Italian Supreme Pizza	1936	Supreme
	The Spicy Italian Pizza	1870.25	Supreme
	The Sicilian Pizza	1813.25	Supreme
	The Four Cheese Pizza	1549.000000000000016	Veggie
	The Five Cheese Pizza	1517	Veggie
	The Mexicana Pizza	1399.75	Veggie